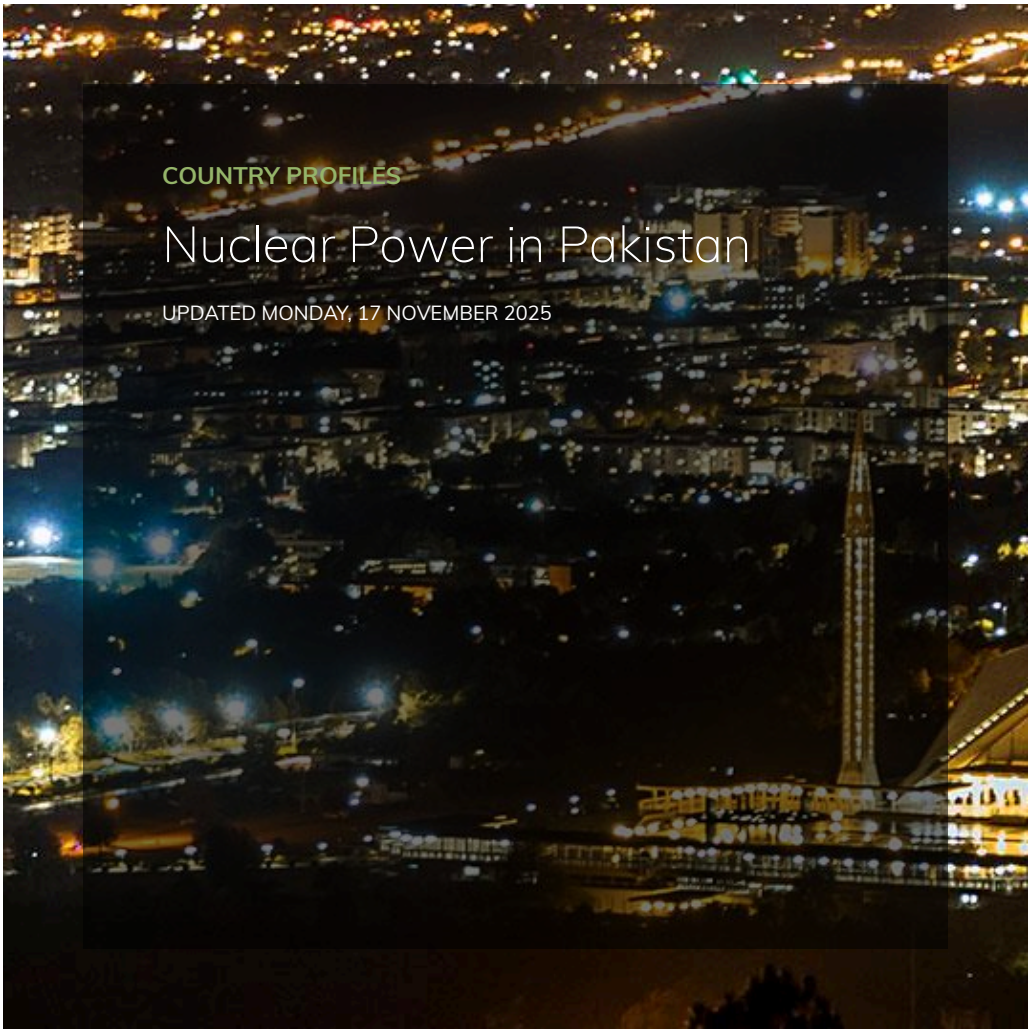


- Pakistan has six operable reactors supplied by China, and a further unit under construction.
- Pakistan's nuclear weapons capabilities have arisen independently of its civil nuclear fuel cycle, using indigenous uranium.
- Because Pakistan is outside the Nuclear Non-Proliferation Treaty due to its weapons programme, it is largely excluded from trade in nuclear plant or materials, which hinders its development of civil nuclear energy. However, China is positive about nuclear

Privacy - Terms

cooperation with Pakistan, and a 2018 International Atomic Energy Agency programme further supports civil nuclear power.

6

OPERABLE REACTORS

3,262 MWe

1

REACTORS UNDER CONSTRUCTION

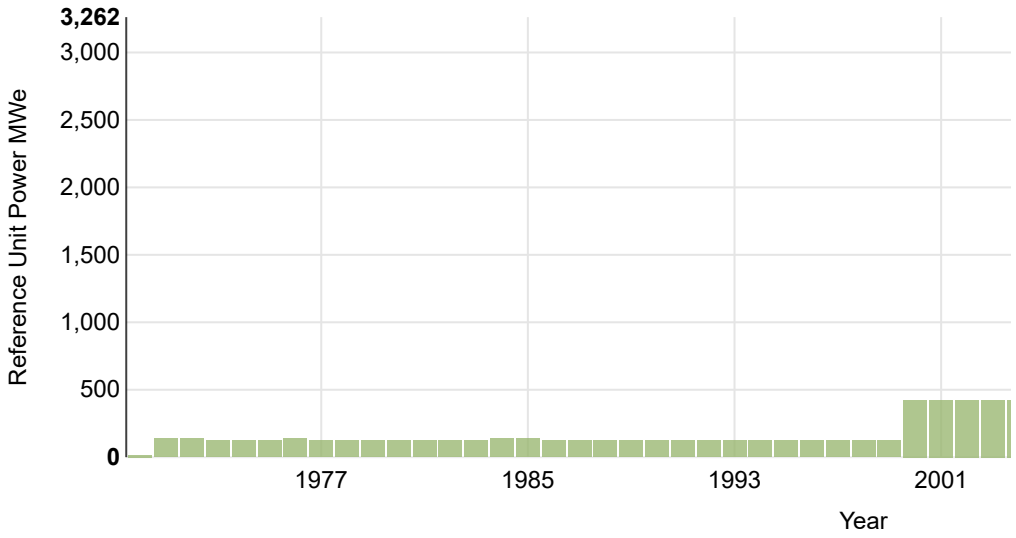
1,100 MWe

1

REACTORS SHUTDOWN

91 MWe

Operable nuclear power capacity



Electricity sector

Total generation (in 2023): 156 TWh

Generation mix: natural gas 41.9 TWh (27.7%); hydropower 40.5 TWh (26.8%); nuclear 26.2 TWh (17.3%); coal 19.3 TWh (12.8%); oil 17.2 TWh (11.4%); wind 4.1 TWh (2.7%); solar 1.1 TWh (0.7%); biofuels & waste 0.8 TWh.

Import/export balance: 0.5 TWh imports, no exports

Total consumption: 117 TWh

Per capita consumption: c. 500 kWh in 2023

Source: International Energy Agency and The World Bank. Data for year 2023.

Nuclear Power Industry

Reactors operating in Pakistan

Reactor Name	Alternative Name	Model	Reactor Type	Net Capacity (MWe)	Construction Start	First Grid Connection
CHASNUPP-1	Chashma Nuclear Power Plant 1	CNP-300	PWR	300	1993-08	2000-06
CHASNUPP-2	Chashma Nuclear Power Plant 2	CNP-300	PWR	300	2005-12	2011-03
Chashma Nuclear Power Plant Unit 3	Chashma Nuclear Power Plant 3	CNP-300	PWR	315	2011-05	2016-10
Chashma Nuclear Power Plant Unit 4	Chashma Nuclear Power Plant 4	CNP-300	PWR	313	2011-12	2017-06
K-2	Karachi 2	HPR1000	PWR	1,017	2015-08	2021-03
K-3	Karachi 3	HPR1000	PWR	1,017	2016-05	2022-03

Karachi is also known as KANUPP; Chashma as CHASNUPP.

Enriched fuel for the PWRs is imported from China.

PAEC is responsible for all nuclear energy and research applications in the country. It has two divisions which are responsible for nuclear power programs: Nuclear Power Generation (NUPG) and Nuclear Power Projects (NUPP). The NUPG directorate oversees the operational units, and the NUPP directorate is concerned with the design and construction of planned units, and is closely aligned with the PNRA.

Karachi 1

PAEC's first nuclear power reactor, Karachi 1 (K1, KANUPP 1) at Paradise Point in Sindh province, about 25 km west of Karachi, was a small 100 MWe (90 MWe net) Canadian pressurized heavy water reactor (PHWR). The unit started up in 1971 and was shut down in August 2021. It was under international safeguards. It was operated at reduced power for several years before its retirement.

At Karachi (KANUPP) a 4800 m^3/day MED desalination plant was commissioned in 2012, though in 2014 it was reported as 1600 m^3/day.

Chashma 1-4

The second unit is Chashma 1 (CHASNUPP 1) in Punjab province in the north, a 325 MWe (300 MWe net) two-loop pressurised water reactor (PWR) supplied by CNNC under safeguards. The main part of the plant was designed by Shanghai Nuclear Engineering Research and Design Institute (SNERDI), based on Qinshan 1. It started up in May 2000. Design lifetime is 40 years. It, and the following three units at the same site, were built using international design codes and standards.

Construction of its twin, Chashma 2 (CHASNUPP 2), started in December 2005. It was reported to cost PKR 51.46 billion ($ 490 million, with $20 million of this financed by China). A safeguards agreement with the IAEA was signed in 2006 and grid connection was in March 2011, with commercial operation in May. Upgrades have added 5 MWe since (to 330 MWe gross).

In June 2008 the government announced plans to build units 3&4 at Chashma, each 320 MWe gross and largely financed by China. A further agreement for China's help with the project was signed in October 2008, and given prominence as a counter to the US-India agreement shortly preceding it.

In March 2009 China's SNERDI announced that it was proceeding with design of Chashma 3&4, with China Zhongyuan Engineering Co. Ltd (CZEC) as the general contractor and China Nuclear Industry No.5 Construction Company as installer. In April 2009, a design contract with SNERDI was signed, and the government said that it had approved the project at a cost of $2.37 billion, with $1.75 billion of this involving "a foreign exchange component". In March 2010 Pakistan announced that it had agreed the terms for Chashma 3&4, whereby China would provide 82% of the total $1.912 billion financing as three 20-year low-interest loans. It would also provide fuel for the reactors' lifetime nominally of 40 years.

The main construction contract was signed in June 2010, detailing that the two 340 MWe CNP-300 (315 MWe net) units were to be completed in eight years. They will have a design lifetime of 40 years and be under IAEA safeguards. Construction of unit 3 officially started at the end of May 2011, and unit 4 in December 2011. Early in 2014 PAEC said they were several months ahead of schedule. In 2015 CZEC said completion of unit 3 would be in 2016, and in fact it was grid-connected in October, with full power and commercial operation in December. Unit 4 started up in March 2017 and was grid-connected late in June 2017.

The Nuclear Suppliers Group (NSG) has raised some questions about China's supply of Chasma 3&4. Contracts for units 1&2 were signed in 1990 and 2000 respectively, before 2004 when China joined the NSG, which maintains an embargo on sales of nuclear equipment to Pakistan. China argued that units 3&4 are similarly 'grandfathered', and arrangements are consistent with those for units 1&2.

During the inauguration of the Chashma 4 unit, the prime minister said the government "is committed to achieve [its] goal of adding 8800 MWe of nuclear energy to the national grid by 2030." This includes Karachi 2, connected to the grid in March 2021, and Karachi 3 currently under construction.

Karachi 2&3

In June 2013 the Planning Commission said that two CNNC 1000 MWe class reactors would be used for Karachi 2&3 (KANUPP 2&3) near Karachi unit 1. Two coastal sites had been under consideration for the twin 1100 MWe units. CNNC in April 2013 announced an export agreement for the ACP1000, nominally 1100 MWe, apparently for Pakistan. This was confirmed in June by the PAEC which said that the next nuclear project would be 1100 MWe class units at the Karachi Coastal power station.

In July 2013 ECNEC approved two units of the Karachi Costal power project with net generation capacity of 2117 MWe. The total cost of this was estimated at PKR 959 billion ($9.116 billion), with $6.5 billion (68%) being vendor finance. PAEC also said that 82% of the total cost would be financed by China. At the end of August 2013 contracts were signed in Shanghai with CNNC, CZEC, China Nuclear Power Engineering Co. Ltd. (CNPE), Nuclear Power Institute of China (NPIC), and East China Electric Power Designing Institute (ECEPDI). Groundbreaking at the site near Paradise Point, 25 km west of Karachi, took place in November 2013, but in October 2014 the Sindh high court ruling stopped site work following a challenge on environmental grounds, and the restraining order was extended to early December. The project was re-launched in August 2015, and construction of the first unit started then.

The Pakistan Nuclear Regulatory Authority received the safety analysis of China’s ACP1000 reactor from CNNC and after completing the review granted a construction licence, for the CNNC version of the Hualong One design, 1161 MWe gross.

In April 2015 China Nuclear Engineering & Construction Group Co (CNEC) won the tender for civil engineering construction and installation work for the conventional island of the plant, which it

said would use Hualong One reactors. Construction of the first unit started in August 2015 and is expected to take 72 months (52 months for the conventional island). Construction of the second unit started at the end of May 2016, according to the IAEA, but without any announcement or notification on the PAEC website. In July 2017 the reactor vessel for unit 2 completed pressure tests at China First Heavy Machinery Group's factory in China, and in September it was installed.

A press report in January 2017 said that work on both units was intensifying to meet the operational target, and that it was a CPEC project. In March 2017 the IAEA approved Pakistan's request to apply international safeguards to both units.

In November 2019, CNNC reported that cold testing had begun for Karachi 2. Hot testing was completed in September 2020, and fuel loading commenced in December 2020. The unit achieved first criticality in February 2021 and was connected to the grid in March, after 67 months construction. The unit began commercial operation in May 2021.

Fuel loading for unit 3 was completed in December 2021. It achieved first criticality in February 2022, and commercial operation in April that year. An operating licence was granted in June 2023.

In light of Pakistan's inability to buy uranium on the open market, CNNC has agreed to provide lifetime fuel supply for the reactors, specified as 60 years.

Chashma 5

In November 2010 the PAEC signed a construction agreement with CNNC for a fifth unit at Chashma. In February 2013 a further agreement was signed by PAEC with CNNC for a 1000 MWe unit at Chashma. Early in 2013 CNNC said that the reactor would be an ACP1000 unit, though not necessarily at Chashma. In November 2017 CNNC signed a cooperation agreement with the PAEC on the construction of Chashma 5 as a Hualong One unit. In June 2023 CNNC announced that it signed the final contracts for the construction of a new Hualong One at Chashma with Pakistan's government. CNNC would build the reactor and provide 85% of the estimated $4.8 billion required for the construction. The ability of Pakistan's government to fund the other 15% hinged on additional taxation measures, rationalizing internal expenditure, and hiking electricity tariffs.

In July 2023 Pakistan's prime minister took part in a ground-breaking ceremony for Chashma 5, and two weeks later the

country's Executive Committee of the National Economic Council gave its approval for the project.

In December 2024 first concrete for unit 5 was poured following regulatory approval earlier that month.

Energy policy

In July 2013 the Executive Committee of the National Economic Council (ECNEC) approved about 3.5 GWe of new power projects totalling PKR 1303 billion ($12.4 billion), comprising 2200 MWe nuclear, 425 MWe gas combined cycle, and 969 MWe hydro. These are designed to reduce the high reliance on oil and to reduce power costs. All depend on Chinese support.

Electricity infrastructure is a significant part of the $51 billion China-Pakistan Economic Corridor (CPEC) projects from 2016 which will link Kashgar in China's Xinjiang region to Pakistan's deep-water port of Gwadar on the Arabian Sea. The CPEC includes roads and railways running much of the length of Pakistan and funded by three Chinese banks, and a 4000 MW HVDC grid development costing $1.5 billion over 2017-18. Some $33 billion of the CPEC total is for energy infrastructure, notably 10 GWe of generating capacity by 2020, mostly coal-fired, which is expected to provide 24% of the country's power by 2020. Lignite is the main fuel envisaged, from the Thar Desert region of Sindh.

CPEC projects are a significant element in China's Belt and Road Initiative (BRI), and are unprecedented in scale in Pakistan.

In 2005 the Energy Security Plan was adopted by the government, which called for a huge increase in generating capacity to more than 160 GWe by 2030. Significant power shortages are reported, and load shedding is common.

Nuclear policy

An expansion of nuclear power capacity has long been a central element of Pakistan's energy policy.

The 2005 Energy Security Plan included the intention of lifting nuclear capacity to 8800 MWe in the long term, 900 MWe of this by 2015 and a further 1500 MWe by 2020. Projections included four further Chinese reactors of 300 MWe each and seven of 1000 MWe, all PWR. There were tentative plans for China to build two 1000 MWe PWR units at Karachi as KANUPP 2&3, but China then

in 2007 deferred development of its CNP-1000 type which would have been the only one of that size able to be exported. Pakistan then turned its attention to building smaller units with higher local content. However, in 2013 China revived its 1000 MWe designs with export intent, and made overtures to Pakistan for the ACP1000 design, which became Hualong One – see below.

In August 2011 it was reported that Pakistan aimed for 8000 MWe nuclear at ten sites by 2030. PAEC has apparently selected six new sites on the basis of Pakistan Nuclear Regulatory Authority (PNRA) and International Atomic Energy Agency (IAEA) advice. These are: the Qadirabad-Bulloki (QB) link canal near Qadirabad Headworks; Dera Ghazi Khan canal near Taunsa Barrage; Taunsa-Panjnad canal near Multan; Nara canal near Sukkur; Pat Feeder canal near Guddu; and Kabul River near Nowshera. Early in 2012 Pakistan Atomic Energy Commission (PAEC) said that four reactors were planned for the Taunsa-Panjnad canal near Multan in Punjab.

In January 2014 PAEC announced its intention to build five further 1100 MWe nuclear reactors to meet anticipated electricity demand, and have 8.9 GWe of nuclear capacity online by 2030. "With more than 55 reactor-years of successful operating experience to its credit, the PAEC can confidently move from technology acquisition status to actually starting contributing sizeable electrical energy to the system." PAEC was then quoted as saying that eight sites would be chosen for a further 32 units, four 1100 MWe units at each, so that nuclear power supplied one-quarter of the country's electricity from 40 GWe of capacity. This evidently assumes a more than tenfold increase in electricity demand by a future date well beyond 2030.

PAEC said an initial 1100 MWe plant would be built at Muzaffargarh, on the Taunsa-Panjnad canal near Multan in southwest Punjab. It was also reported that discussions with China were under way to supply three nuclear power units for about $13 billion.

Despite being outside the UN's Nuclear Non-Proliferation Treaty (NPT), a four-year technical cooperation project – "Strengthening and Enhancing Capabilities of Pakistan's National Institutions to Support a Safe, Reliable and Sustainable Nuclear Power Programme," referred to as PAK2007 – was launched in 2018. The IAEA amalgamated four of its pre-existing national technical cooperation projects, which supported regulators, operators, waste managers and non-destructive testers to bring together all the relevant nuclear power stakeholders. This considerably boosted

international cooperation with Pakistan's nuclear power programme.

Front cycle

Front end

The government set a target of producing 300 tonnes of uranium per year from 2015 to meet one-third of anticipated requirements, but this has not been realised. Low-grade ore is known in central Punjab at Bannu Basin and Suleman Range. In 2015 production was 45 tU.

In July 2017 CNNC signed a framework agreement with PAEC for technical cooperation in the exploration and development of uranium resources.

A small (15,000 SWU/yr) uranium centrifuge enrichment plant at the Khan Research Laboratories (KRL) in Kahuta has been operated since 1984 and does not have any apparent civil use. It was expanded threefold in about 1991, and further since then. A newer plant not under safeguards is reported to be at KRL. It is not clear whether PAEC has any involvement with these plants.

Enriched fuel for the PWRs is imported from China.

In 2006 PAEC announced that it was preparing to set up separate and purely civil conversion, enrichment and fuel fabrication plants as a new $1.2 billion Pakistan Nuclear Power Fuel Complex (PNPFC) for PWR-type reactors which would be under IAEA safeguards and managed separately from existing facilities. However, constraints imposed on Pakistan by the Nuclear Suppliers Group may mean that all civil nuclear development is tied to China, and there may be no point in proceeding with this project.

Waste management

The PAEC has responsibility for radioactive waste management. A Central Radioactive Waste Management Fund is proposed in a new policy. Waste management centres are proposed for Karachi and Chashma.

Used fuel is currently stored at each reactor in pools. Longer-term dry storage at each site is proposed. The question of future

reprocessing remains open.

Research and development

The Pakistan Institute of Nuclear Science & Technology (PINSTECH) at Rawalpindi near Islamabad is managed by the PAEC and is one of the largest science and technology research establishments in the country. It has conducted research into reprocessing used nuclear fuel, though today it claims to be focused on research in medicine, biology, materials and physics, including production of medical radioisotopes.

Pakistan has a 10 MW pool-type research reactor, PARR-1, of 1965 vintage, supplied by the USA under the Atoms for Peace program. It was converted to use low-enriched uranium fuel in 1991, and upgraded from 5 to 10 MW. PARR-2 is an indigenous 30 kW miniature neutron source reactor (MNSR) based on Chinese design and using high-enriched fuel operating since 1974. Both are located at the PINSTECH Laboratory, Nilore, near Islamabad. They are under IAEA safeguards. One of them produces some Mo-99 from HEU targets.

New Labs at PINSTECH in Rawalpindi is reported to be a reprocessing plant for weapons-grade plutonium production, and not under safeguards. It is run by PAEC and operational since 1981. This was apparently the culmination of a plutonium weapons program predating the Kahuta HEU weapons program, and replaced an unfinished much larger reprocessing plant (100 t/yr) being built at Chashma by France, but cancelled in 1978.

At Khushab, 200 km south of Islamabad, there are four heavy water reactors dedicated to production of weapons-grade plutonium, plus a heavy water plant. The first of these, a 'multipurpose' PHWR estimated at 30-40 MWt, started operating in 1998. Then a larger (40-50 MWt) heavy water reactor was built there from about 2002, and appeared to be operational at the end of 2009. In 2006, construction started on a third reactor, similar to and adjacent to the second, and this appeared to be operational by the end of 2013. A similar but larger (90 MWt) fourth reactor was built from 2011 a few hundred metres away, and appeared to be operational in January 2015. These seem to add up to a substantial plutonium production capacity. Khushab is reported to be making demands upon the country's limited uranium resources. A small heavy water plant is nearby.

Reprocessing of military material is reported to take place at Chashma, 80 km west, and the original French reprocessing plant is apparently under renewed construction there, a couple of kilometres southwest of Chashma 1-4 power reactors.

The Khan Research Laboratories (KRL) at Kahuta in Punjab is described as a weapons engineering R&D institute and research laboratory, focused on producing high-enriched uranium using centrifuge technology originally stolen from Urenco by Dr Abdul Q Khan. Set up about 1976 as the Engineering Research Laboratories it was a key part of Pakistan's weapons program, supported by the Army Corps of Engineers in competition with the plutonium program being pursued by PAEC. It was renamed in honour of Dr Khan in 1981.

Regulation

The Pakistan Nuclear Regulatory Authority (PNRA) is responsible for licensing and supervision, and regulates the safety and security of all civil nuclear materials and facilities. In respect to the Chashma reactors, and presumably also the Karachi Coastal power project, it works closely with China's National Nuclear Safety Administration. It was formed in 2001, superseding the Pakistan Nuclear Regulatory Board (set up by PAEC) and the Directorate of Nuclear Safety and Radiation Protection.

Pakistan is party to the Convention on Nuclear Safety and two international conventions for early notification and assistance.

Non-proliferation

Pakistan is not party to the Nuclear Non-Proliferation Treaty (NPT) but does have its civil power reactors and two research reactors (PARR 1&2) under item-specific IAEA safeguards. An agreement for two further 340 MWe reactors – Chashma 3&4 – came into force in April 2011. In March 2017 the IAEA approved Pakistan’s request to apply international safeguards to both Karachi Coastal units, and the agreement on this came into force in May. Pakistan has refused calls for international inspections of its enrichment activities.

The Institute for Science and International Security (ISIS) said in May 2018: "Unlike India, Pakistan has barely separated its civil and military nuclear facilities and in general remains highly secretive about its nuclear program. The opaqueness of Pakistan's nuclear program, its expanding nuclear weapons arsenal, and its refusal to

separate its military and civilian nuclear program are cited as reasons by many countries opposing Pakistan's membership of the Nuclear Suppliers Group (NSG), as well as [opposing] any country supplying it with additional nuclear power reactors or other fuel cycle facilities or capabilities. Such supply is viewed by many countries as a violation of NSG guidelines."

Pakistan's Kahuta project (incorporating Project-706) to produce a uranium bomb was launched in 1972, following a disastrous war with India. It was partly financed by Libya to 1979. In May 1974 India exploded a nuclear test close to the Pakistan border, galvanising Pakistani efforts. The project was disbanded in 1983 after a successful cold test of weapons components.

In May 1998 Pakistan exploded five atomic devices in Baluchistan. At least one was evidently made from enriched uranium, but the Chagai II test in Kharan desert used plutonium produced by New Labs.

Pakistan is reported to be the sole nation blocking agreement of the Fissile Material Cutoff Treaty (FMCT) in Geneva negotiations.

Through the activities of Dr Khan, a centrifuge plant and nuclear weapons designs were secretly supplied to Libya from the late 1990s to 2003 to help build a weapons program there. He also transferred centrifuge technology to North Korea in the 1990s, and to Iran. This is the main basis for the NSG refusing to ease nuclear trade sanctions for Pakistan, as it has for India. China is the only country to act in defiance of trade sanctions, and has deepened cooperation since the international US-led concessions to India in 2008. This is most obvious in 2013 agreements to build the twin-unit Karachi Coastal power plant and the CNNC contract with PAEC for lifetime fuel supply for this.

Addressing the 3rd Nuclear Security Summit at The Hague in March 2014, the prime minister said that Pakistan had been running a safe and secure nuclear program for over four decades with the expertise, manpower and infrastructure to produce civil nuclear energy. He called for Pakistan’s inclusion in all international export control regimes, especially the NSG. He pointed out that international treaties and forums would supplement Pakistan’s national actions to fortify nuclear security.

Domestically, he said that today the country’s nuclear security is supported by five pillars – a strong command and control system led by the National Command Authority (NCA); an integrated intelligence system; a rigorous regulatory regime; a comprehensive

export control regime; and active international cooperation. The security regime covers physical protection, material control and accounting, border controls and radiological emergencies, he said.

Pakistan is a major recipient of technical cooperation from the IAEA, and is one of 35 members of the IAEA Board of Governors, though it remains outside the NPT.

Appendix: Nuclear Proliferation Issues

This appendix is based on paper by Michael Wilson, 1995, The Nuclear Future: Asia and Australia and the 1995 Conference on Non-Proliferation, publ. by Griffith University. Used with author's permission.

Pakistan (along with India and Israel) was originally a "threshold" country in terms of the international non-proliferation regime (see page on Safeguards to Prevent Nuclear Proliferation), possessing, or quickly capable of assembling one or more nuclear weapons. Their nuclear weapons capability at the technological level was recognised (all have research reactors at least) along with their military ambitions. Then in 1998 India and Pakistan's military capability became more overt. All three remained outside the 1970 Nuclear Non-Proliferation Treaty (NPT), which 186 nations have now signed. This led to their being largely excluded from trade in nuclear plant or materials, except for safety-related devices for a few safeguarded facilities.

Regional rivalry

Relations between Pakistan and India are tense and hostile, and the risks of nuclear conflict between them have long been considered quite high.

In 1974 India exploded a "peaceful" nuclear device and then in May 1998 India and Pakistan each exploded several nuclear devices underground. This heightened concerns regarding an arms race between them.

Kashmir is a prime cause of bilateral tension, its sovereignty being in dispute since 1948. There is persistent low-level military conflict due to Pakistan backing a Muslim rebellion there.

Both countries engaged in a conventional arms race in the 1980s, including sophisticated technology and equipment capable of delivering nuclear weapons. In the 1990s the arms race quickened. In 1994 India reversed a four-year trend of reduced allocations for defence, and despite its much smaller economy, Pakistan pushed its own expenditures yet higher. Both then lost their patrons: India, the former USSR, and Pakistan, the USA.

Pakistan has offered to disarm and join the NPT if India would, although both countries see the NPT as unfair and India would prefer other international arrangements for limiting weapons proliferation.

Pakistan's weapons technology is based on the production of highly enriched uranium suitable for nuclear weapons, utilising indigenous uranium. It has at least one small centrifuge enrichment plant. In 1990 the US administration cut off aid because it was unable to certify that Pakistan was not pursuing a policy of manufacturing nuclear weapons though this was relaxed late in 2001. In 1996 the USA froze export loans to China because it was allegedly supplying centrifuge enrichment technology to Pakistan.

Pakistan made it clear since early 1996 that if India staged a nuclear test, it had done the basic development work and would immediately start assembling its own nuclear explosive device. Its nuclear weapons capability has since been demonstrated, and it is assumed to have enough highly-enriched uranium for up to 40 nuclear warheads.

In April 1998 Pakistan test fired a long-range missile capable of reaching Madras in southern India, pushing home the point by naming it after a 12th Century Muslim conqueror. This development diminished India's military advantage over Pakistan.

Since then Pakistan has been exposed as having supplied sensitive nuclear technology, notably centrifuge enrichment designs and equipment, to Iran, Libya and North Korea. Its non-proliferation credentials therefore stand in stark contrast to India's. Pakistan's security concerns derive from:

- India's possession of a nuclear weapons capability.
- Its development of short and intermediate-range missiles and, since their partition in 1947.
- Its defeat by India in two of three wars, notably in East Bengal, now Bangladesh, in 1972.

Nuclear arms control in the region

The public stance of Pakistan and India on non-proliferation differs markedly.

Pakistan has initiated a series of regional security proposals. It has repeatedly proposed a nuclear-free zone in South Asia and has proclaimed its willingness to engage in nuclear disarmament and to sign the NPT if India would do so. This would involve disarming and joining as non-weapons states. It has endorsed a US proposal for a regional five power conference to consider non-proliferation in South Asia

India has taken the view that solutions to regional security issues should be found at the international rather than the regional level, since its chief concern is with China. It therefore rejects Pakistan's proposals.

Instead, India's 'Gandhi Plan', put forward in 1988, proposed the revision of the NPT, which it regards as inherently discriminatory in favour of the five nuclear-weapons states, and a timetable for complete nuclear weapons disarmament. It endorsed early proposals for a Comprehensive Test Ban Treaty (CTBT) and for an international convention to ban the production of highly enriched uranium and plutonium for weapons purposes, known as the 'cut-off' convention.

The USA has, for some years pursued a variety of initiatives to persuade India and Pakistan to abandon their nuclear weapons programs and to accept comprehensive international safeguards on all their nuclear activities. To this end the Clinton administration proposed a conference of nine states, comprising the five established nuclear-weapons states, along with Japan, Germany, India and Pakistan.

This and previous similar proposals have been rejected by India, which countered with demands that other potential weapons states, such as Iran and North Korea, should be invited, and that regional limitations would only be acceptable if they were accepted equally by China. The USA would not accept the participation of Iran and North Korea and such initiatives lapsed.

Another, more recent approach, centres on the concept of containment, designed to 'cap' the production of fissile material for weapons purposes, which would hopefully be followed by 'roll back'. To this end India and the USA jointly sponsored a UN General

Assembly resolution in 1993 calling for negotiations for a 'cut-off' convention. Should India and Pakistan join such a convention, they would have to agree to halt the production of fissile materials for weapons and to accept international verification on their relevant nuclear facilities (enrichment and reprocessing). In short, their weapons programs would be thus 'capped'. It appeared that India was prepared to join negotiations regarding such a Fissile Material Cut-off Treaty (FMCT) under the UN Conference on Disarmament (UNCD).

However, despite the widespread international support for a FMCT, formal negotiations on cut-off have yet to begin. The UNCD can only approve decisions by consensus and since the summer of 1995, the insistence of a few states to link FMCT negotiations to other nuclear disarmament issues has brought progress on the cut-off treaty there to a standstill. In connection with its 2006 agreement with the USA, India has reiterated its support for a FMCT.

Bilateral confidence-building measures between India and Pakistan to reduce the prospects of confrontation have been limited. In 1990 each side ratified a treaty not to attack the other's nuclear installations, and at the end of 1991 they provided one another with a list showing the location of all their nuclear plants, even though the respective lists were regarded as not being wholly accurate. Early in 1994 India proposed a bilateral agreement for a 'no first use' of nuclear weapons and an extension of the 'no attack' treaty to cover civilian and industrial targets as well as nuclear installations.

Having promoted the CTBT since 1954, India dropped its support in 1995 and in 1996 attempted to block the treaty. Following the 1998 tests the question has been reopened and both Pakistan and India have indicated their intention to sign the CTBT. Indian ratification may be conditional upon the five weapons states agreeing to specific reductions in nuclear arsenals.

Notes & references

General sources

Pakistan Atomic Energy Commission (PAEC)

Related information

India
Safeguards to Prevent Nuclear Proliferation